Conformal Winner Sets: A First Evaluation

2026-11-06

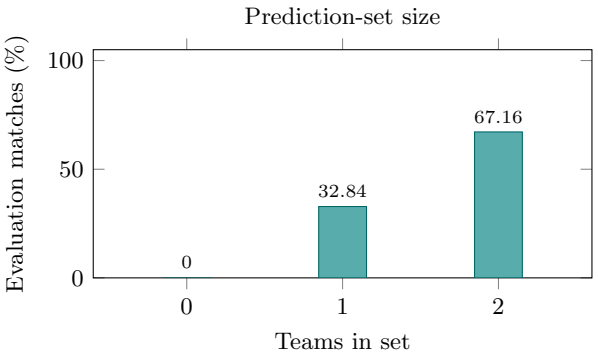
1 Setup

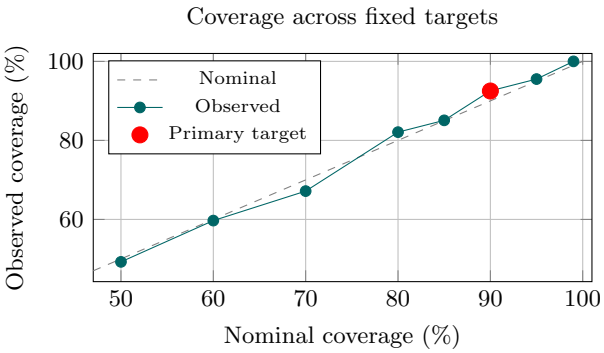
I took the finished model, and the unused test matches, splitting it 67 calibration matches, 67 evaluation matches. I took a single nonterminal prefix per match to evaluate.

2 Method

I used split conformal prediction with score $s_i = 1 - p(y_i \mid x_i)$ and the $k = \lceil (n+1)(1-\alpha) \rceil$th ordered calibration score q. If $k > n$, both teams are included. I used $\alpha = 0.1$. See Angelopoulos and Bates for the method I used.

Statistic	Evaluation result
Nominal / observed coverage	90% / 92.5%
Observed coverage, descriptive 95% Wilson interval	[83.7%, 96.8%]
Quantile rank / calibration count; threshold q	62 / 67; 0.665159
Mean prediction-set size (maximum 2)	1.672
Empty / singleton / two-team sets	0 / 22 / 45
Accuracy among singleton sets	77.3%
Original model's argmax accuracy	59.7%





AI-Generated summary table.

3 Results

At the 90% target, the sets contained the winner in 62 of 67 evaluation matches (92.5%). it had one team in 22 matches and both in 45. Accuracy among singletons is an informative statistic, it tells us about the confidence of the model.

4 Conclusion

I would like to try using more matches, and see how it improves. Also, some of these matches may have shared players or been across different patches, so that may hurt the gaurantees of exchangability. I also want to check how these set size plots change over time, as it should be a monotonic decrease in set size over time.